LAB 2: Cloud Infrastructure & Architecture

Objective

To understand the basic components of cloud infrastructure and learn how cloud architecture works in modern cloud computing environments.

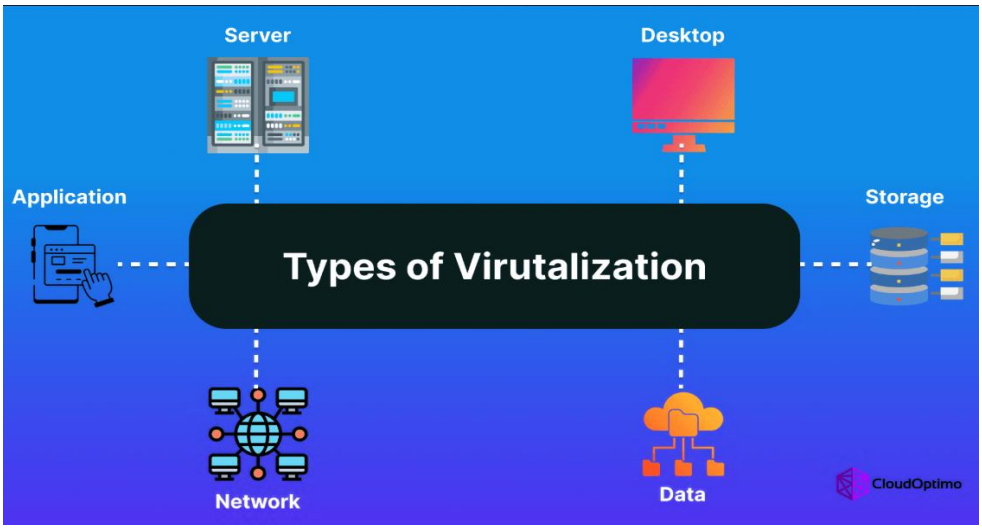
Introduction

Cloud infrastructure is the collection of hardware and software components used to provide cloud computing services. It includes servers, storage, networking devices, virtualization software, and management tools that deliver computing resources through the internet.

Cloud architecture describes how these components are organized and connected to provide scalable, reliable, and accessible services. It allows organizations to run applications and store data without managing physical hardware.

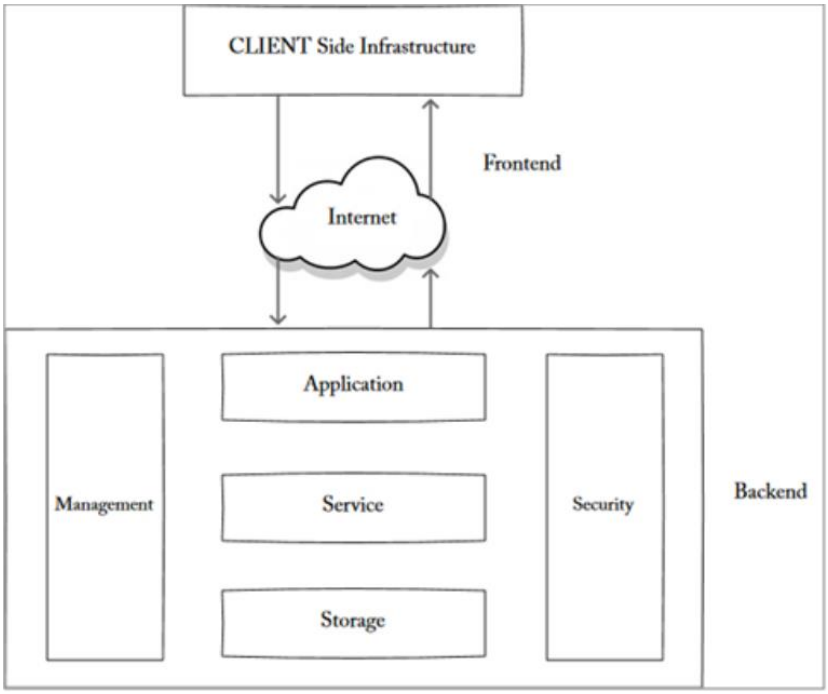
Main Components of Cloud Infrastructure

1. **Compute (Servers)**
   Provides processing power to run applications and services.
2. **Storage**
   Stores files, databases, and application data in the cloud.
3. **Networking**
   Connects cloud resources and allows communication through the internet.
4. **Virtualization**
   Allows multiple virtual machines to run on a single physical server.
5. **Management Software**
   Tools used to monitor, control, and manage cloud resources.



Basic Cloud Architecture Layers

1. **Front-End Layer**
   The user side, where clients access cloud services using browsers or applications.
2. **Back-End Layer**
   The cloud provider side that contains servers, storage, databases, and management systems.
3. **Network Layer**
   Connects the front-end and back-end through the internet.



Lab Tasks:

Task 01: Draw a simple diagram of cloud infrastructure showing client, internet, and cloud services.

Task 02: Label at least **3 components** and write their function (e.g., compute, storage, network).

Physical Infrastructure Identification Activity:

Goal: Connect cloud with real-world hardware

1. Look around your computer lab.
2. Identify:
   - One device that represents **Compute**

    - One that represents **Storage**
    - One that represents **Networking**

3. Write what each one does.

Example:

- CPU → Compute
- Hard Disk → Storage